

AISI 1030 Steel, as rolled

Categories: [Metal](#); [Ferrous Metal](#); [Carbon Steel](#); [AISI 1000 Series Steel](#); [Medium Carbon Steel](#)

Material Notes: Moderate strength and hardness in the as-rolled condition. Fair machinability and weldability. Precautions must be taken when welding to avoid cracking during cooling.

Key Words: SAE J414, ASTM A108; ASTM A29;ASTM A510; ASTM A512; ASTM A519; ASTM A544; ASTM A545; ASTM A546; ASTM A576; ASTM A682, DIN 1.1172; FED QQ-S-635 (C1030); FED QQ-S-700 (C1030); MIL SPEC MIL-S-11310 (CS1030); SAE J403; SAE J412

Vendors: No vendors are listed for this material. Please [click here](#) if you are a supplier and would like information on how to add your listing to this material.

Physical Properties	Metric	English	Comments
Density	7.85 g/cc	0.284 lb/in ³	

Mechanical Properties	Metric	English	Comments
Hardness, Brinell	179	179	
Hardness, Knoop	200	200	Converted from Brinell
Hardness, Rockwell B	88	88	Converted from Brinell
Hardness, Vickers	188	188	Converted from Brinell
Tensile Strength, Ultimate	550 MPa	79800 psi	
Tensile Strength, Yield	345 MPa	50000 psi	
Elongation at Break	32 %	32 %	In 50 mm
Reduction of Area	57 %	57 %	
Modulus of Elasticity	206 GPa	29900 ksi	
Bulk Modulus	163 GPa	23600 ksi	Estimated from elastic modulus
Poissons Ratio	0.29	0.29	Typical for steel
Shear Modulus	80.0 GPa	11600 ksi	Estimated from elastic modulus
Izod Impact	75.0 J	55.3 ft-lb	

Electrical Properties	Metric	English	Comments
Electrical Resistivity	0.0000166 ohm-cm @Temperature 20.0 °C	0.0000166 ohm-cm @Temperature 68.0 °F	annealed specimen

Thermal Properties	Metric	English	Comments
CTE, linear	11.7 µm/m-°C @Temperature 20.0 °C	6.50 µin/in-°F @Temperature 68.0 °F	
Specific Heat Capacity	0.486 J/g-°C @Temperature >=100 °C	0.116 BTU/lb-°F @Temperature >=212 °F	annealed
Thermal Conductivity	51.9 W/m-K	360 BTU-in/hr-ft ² -°F	

Component Elements Properties	Metric	English	Comments
Carbon, C	0.27 - 0.34 %	0.27 - 0.34 %	
Iron, Fe	98.67 - 99.13 %	98.67 - 99.13 %	As remainder
Manganese, Mn	0.60 - 0.90 %	0.60 - 0.90 %	
Phosphorus, P	<= 0.040 %	<= 0.040 %	
Sulfur, S	<= 0.050 %	<= 0.050 %	

References for this datasheet.

Some of the values displayed above may have been converted from their original units and/or rounded in order to display the information in a consistent format. Users requiring more precise data for scientific or engineering calculations can click on the property value to see the original value as well as raw conversions to equivalent units. We advise that you only use the original value or one of its raw conversions in your calculations to minimize rounding error. We also ask that you refer to MatWeb's [terms of use](#) regarding this information. [Click here](#) to view all the property values for this datasheet as they were originally entered into MatWeb.